25901119 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOT200, MG4DSEBBT200 - FOOD SCIENCE AND QUALITY CONTROL

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer

Answer any **10** questions

Each question carries **2** marks

1. What role does fermentation play in traditional diets? [K] / [CO1]
2. Identify the mineral that helps in proper muscle contraction. [K] / [CO1]
3. Why are carotenoids more stable than chlorophyll during cooking? [K] / [CO1]
4. Name an adulterant found in poultry products. [U] / [CO2]
5. Mention any two disadvantages of natural sweeteners used in food. [U] / [CO4]
6. Name any two common adulterants found in milk. [U] / [CO2]
7. Give two examples of common food adulterants. [K] / [CO2]
8. What happens to fruits during moisture loss? [K] / [CO2]
9. Describe how smoking acts as a preservation method. [U] / [CO3]
10. Name one control point in quality control. [A] / [CO1, CO3]

11. What does the FSSAI logo indicate? [U] / [CO1, CO3]

12. Name one principle of HACCP. [K] / [CO1, CO3]

[2x10 = 20]

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. Explain the difference between fat soluble and water soluble vitamins with examples. [K] / [CO1]

14. Why are microbial sources preferred for clean label products and large scale industrial applications? [K] / [CO1]

15. Explain the importance of carbohydrates in human nutrition. [K] / [CO1]

16. Explain food adulteration and its impact on human health. [U] / [CO2]

17. Differentiate between stabilizers and thickeners with suitable examples. [U] / [CO1]

18. Explain food poisoning and its common causes. [K] / [CO1, CO2]

19. Define 'Antioxidant Preservatives' and explain two different ways they protect food quality. [K] / [CO3]

20. Classify foods based on susceptibility to spoilage and give suitable examples. [K] / [CO2]

21. Discuss the importance and procedure of inspection in food industries. [An] / [CO1, CO3]

22. Describe the symptoms and control measures of ergotism. [U] / [CO1, CO2]

[5x8 = 40]

Part C

Essay

Answer any **1** questions

Each question carries **10** marks

23. Define lipids. Classify fatty acids into saturated and unsaturated fatty acids with examples and functions. [K] / [CO1]

24. Describe high-temperature preservation methods used in food processing. [K] / [CO3]

[10x1 = 10]